

# Solution

CS-4053 Recommender System

Final Exam

Spring 2025

Data: 2<sup>nd</sup> June 2025

---

Question 1:

- i. b
- ii. b
- iii. b
- iv. d
- v. d

Question 2:

$$a) \quad P(X|r = 1) = \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} = 0.008 \quad P(r = 1) = \frac{0+0.01}{2+0.05} = 0.004$$

$$P(X|r = 3) = \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} = 0.008 \quad P(r = 3) = \frac{0+0.01}{2+0.05} = 0.004$$

$$P(X|r = 5) = \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} = 0.008 \quad P(r = 5) = \frac{0+0.01}{2+0.05} = 0.004$$

$$P(X|r = 2) = \frac{0+0.01}{1+0.05} \times \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{1+0.05} = 0 \quad P(r = 2) = \frac{1+0.01}{2+0.05} = 0.492$$

$$P(X|r = 4) = \frac{0+0.01}{1+0.05} \times \frac{0+0.01}{0+0.05} \times \frac{0+0.01}{0+0.05} = 0 \quad P(r = 4) = \frac{1+0.01}{2+0.05} = 0.492$$

$$P(r = 1|X) = P(r = 3|X) = P(r = 5|X) = 0.00003$$

$$P(r = 2|X) = P(r = 4|X) = 0$$

Either of Item **1**, **3** or **5** will be recommended.

- b) We do not generally need significance weighting with Probabilistic Collaborative Filtering because it is already based on the principle of common ratings. The number of common ratings is already taken into consideration in the basic formulae for prior and likelihood.

Question 3:

- a) This approach is likely to deteriorate the performance of the model. After every decomposition and optimization, a new local loss will be introduced, which would contribute to increasing the global loss.

a)  $Sim(P2, P1) = 0.516$   
 $Sim(P2, P3) = 0.410$

$$\text{Sim}(P2, P4) = 0.031$$

**Page 1** will be recommended based on similarity in features with Page 2.

**Question 4:**

- a)  $\text{Sim}(\text{req}, I1) = 0.927$   
 $\text{Sim}(\text{req}, I2) = 0.788$   
 $\text{Sim}(\text{req}, I3) = 0.769$

**Item 1** will be recommended.

- b) No. It cannot be affected by sparsity, since it only requires the latest values for user and item attributes and no historical interactions.

**Question 5:**

- a) The feature embeddings are following uniform distribution; any other data sample would most likely be producing a completely different data distribution over feature space. The model cannot thus predict the rating/label with as much accuracy.
- b) It cannot produce ratings for multiple items at once because the item and user features are concatenated to become a single training sample, this concatenation would lead to loss of identification information of these items.

c)

```
data = {
    'Item1': [2, 3, nan, 2],
    'Item2': [nan, 1, 4, 5],
    'Item3': [4, 2, 1, 4],
    'Item4': [nan, 5, 3, 3]
}
```

```
interactions = pd.DataFrame(data, index=['U1', 'U2', 'U3', 'U4'])
user1_ratings = interactions.loc['U1'].dropna()
print(interactions.loc['U1'])
```